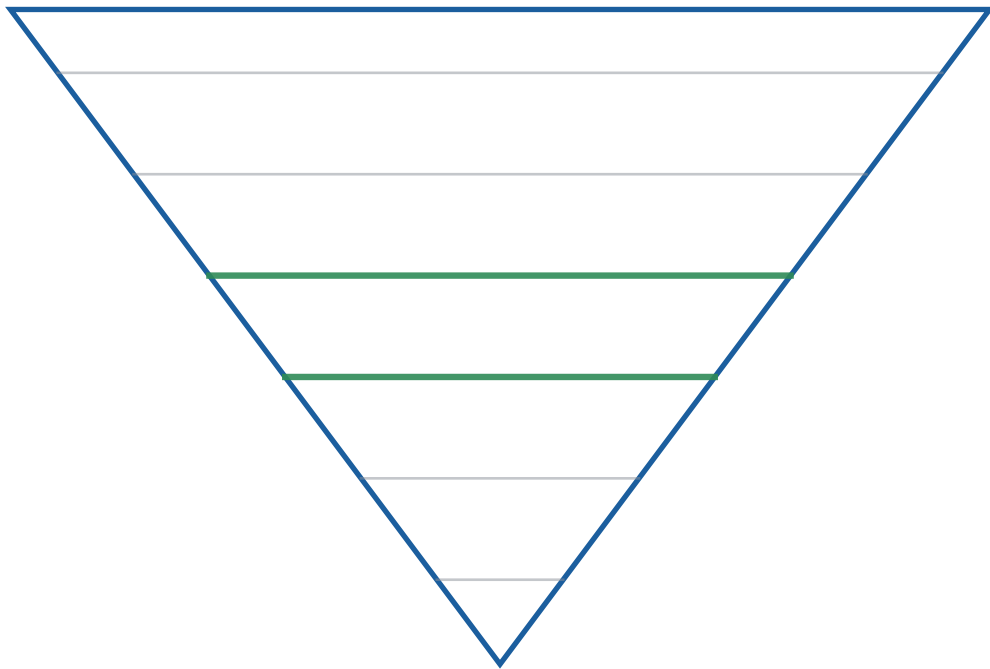


Constraints of Existence

A technical report on the eight laws of the playground world

all organisms that could exist



survivor (PO -> tip)

Gradient-free neuroevolution · model defined by constraints, not parameters

Read the state, not the output · change the world, not the score

1. Executive summary

What this is

GENREG models an organism not as a fixed network but as the INTERSECTION of the laws it must satisfy to keep existing. Each law removes degrees of freedom from the space of possible behaviours; stack enough and a single survivor precipitates. The native complexity metric is therefore PO -- the number of binding constraints (axes) -- not a parameter count.

This report documents the eight constraints implemented in the interactive playground, the behaviour each one selects for, the evidence from direct simulation, and -- explicitly -- where each one is weak, fragile, or still unresolved.

Headline findings

1. Stacking constraints monotonically collapses surviving strategy diversity (Fig 1): the cone is real and measurable.
2. Single-law signatures reproduce cleanly: perception cost shrinks vision; scarcity disperses the population; energy is the precondition for any selection at all.
3. The reproduction operator is itself a load-bearing law. Switching from asexual cloning to mate-based sex changes the EVOLUTIONARY MEANING of communication: under cloning the signal-following gene can drift anti-social; under sex it is floored pro-social, because an organism that avoids others never finds a mate (Fig 8).

Stance

Findings are reported at the strength the data supports. Clean, reproducible effects are stated plainly; seed-sensitive or parameter-sensitive effects are flagged as such; mechanisms we have not yet closed are listed as open questions, not glossed.

Reading guide

Section 2 gives the methods. Sections 3-10 treat one constraint each (mechanism, parameters, evidence, shortcomings). Section 11 covers the reproduction x communication interaction. Section 12 lists cross-cutting limitations and the active research agenda.

2. Methods

Organism and world

A population of organisms lives on a 100x100 continuous board with food patches. Each organism carries a 6-gene genome: four behavioural weights (attraction to food, to remembered food, to/from neighbours, to/from signallers) and two traits (vision radius, movement speed). Every step: sense -> decide a heading -> move -> eat -> pay the costs imposed by active laws -> reproduce.

```
genome = [ w_food, w_mem, w_sep, w_sig, vision, speed ]
```

Evolution (gradient-free)

No gradients are used anywhere. Default reproduction is steady-state: each step the lowest-energy organisms are replaced by mutated copies of high-energy ones, placed near the parent (local / kin reproduction). The Reproduction law (Sec 10) swaps this for mate-based sexual reproduction.

The PO metric

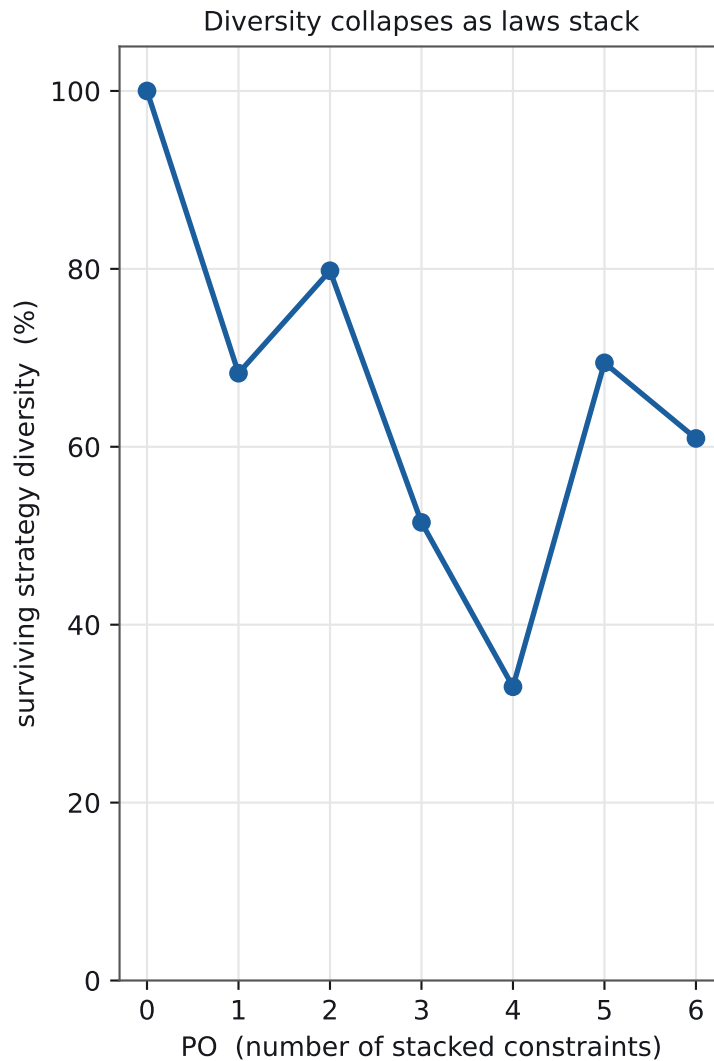
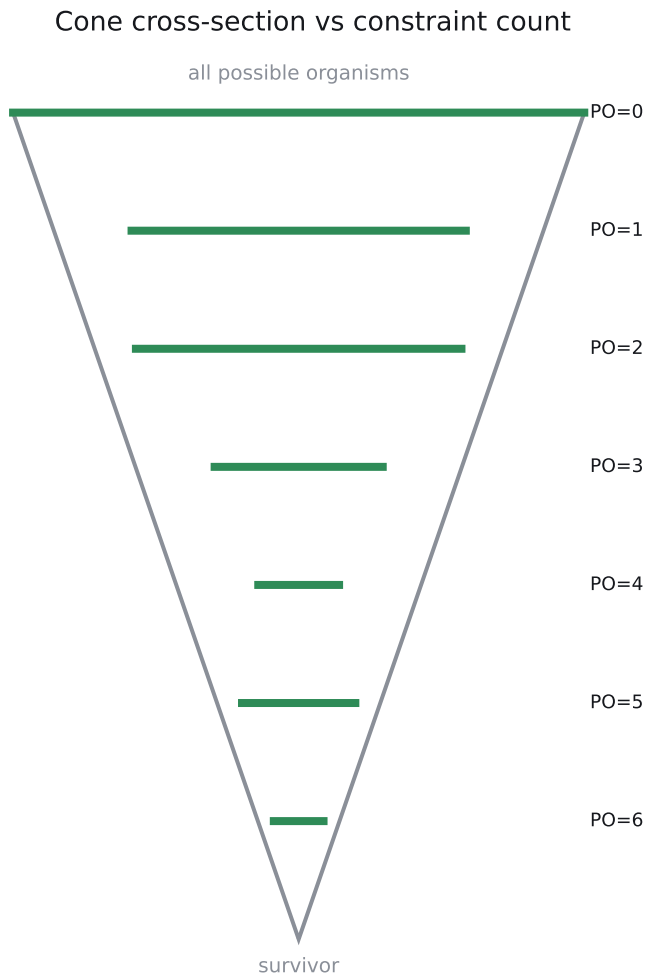
Strategy diversity is measured as the mean relative dispersion of the genome across the population, tanh-compressed, normalised to its value at initialisation (100%). As constraints select harder, surviving genomes converge and this fraction drops -- the cone cross-section.

Measurement

Every figure is generated from direct simulation of this engine (no hand-drawn curves). Unless noted, runs use 32 organisms, 600-900 steps, seed 7. Where a result depends on the random seed, multiple seeds are shown.

Caveat: the engine is deliberately minimal -- a linear genome, one hidden behavioural rule, a heuristic diversity metric. It is an instrument for studying constraint geometry, not a high-fidelity ecology. Absolute numbers are not meaningful; contrasts (law on vs off) are.

Figure 1. The PO cone: each constraint removes degrees of freedom



3. Energy (survival)

COLOUR: red AXIS: survival STATUS: foundational, verified

Definition

Metabolism drains energy every step; an organism that reaches zero is removed. Energy is the precondition that makes selection exist at all.

Mechanism / parameters

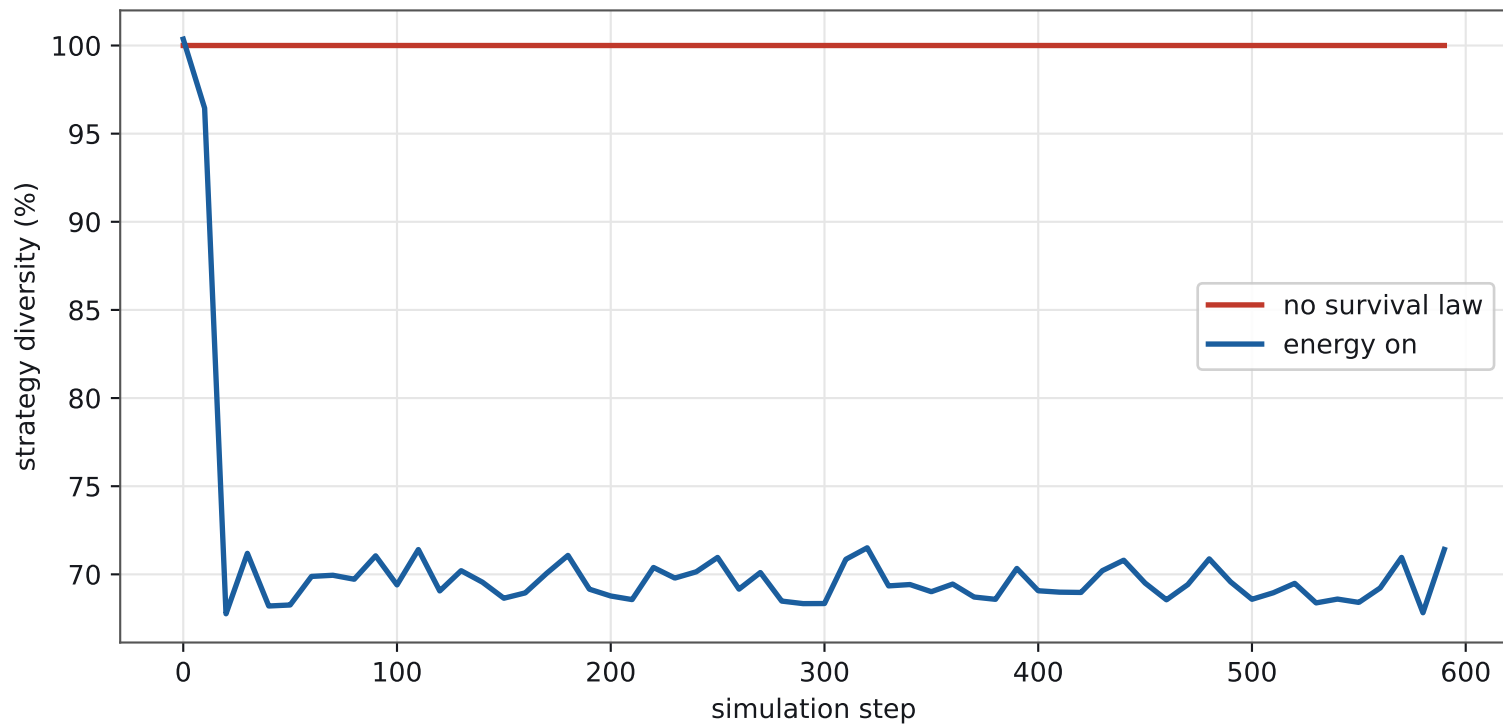
```
energy -= metabolism          (default 0.40 / step)
food raises energy; death at energy <= 0
```

Observed effect

With no survival law there is no selection pressure: the genome drifts and strategy diversity stays at ~100% indefinitely. Switching energy on collapses diversity as fitter foragers displace the rest (Figure 2). Energy is the substrate every other law acts through -- a perception cost or scarcity penalty only matters because it shortens a life.

Open / weak: the fixed steady-state operator means population size is held constant, so 'alive count' can understate die-off (low-energy organisms are recycled). Lifespan, not headcount, is the honest fitness signal. Under very harsh stacks the population can sit pinned at the energy floor while still 'evolving' on relative rank -- a regime where absolute readouts mislead.

Figure 2. Energy is the precondition for selection



4. Time / Occam (movement cost)

COLOUR: purple AXIS: parsimony STATUS: real, mild

Definition

Movement costs energy in proportion to speed. Wasted motion is paid for, so efficient, direct behaviour is selected -- an evolutionary Occam's razor.

Mechanism / parameters

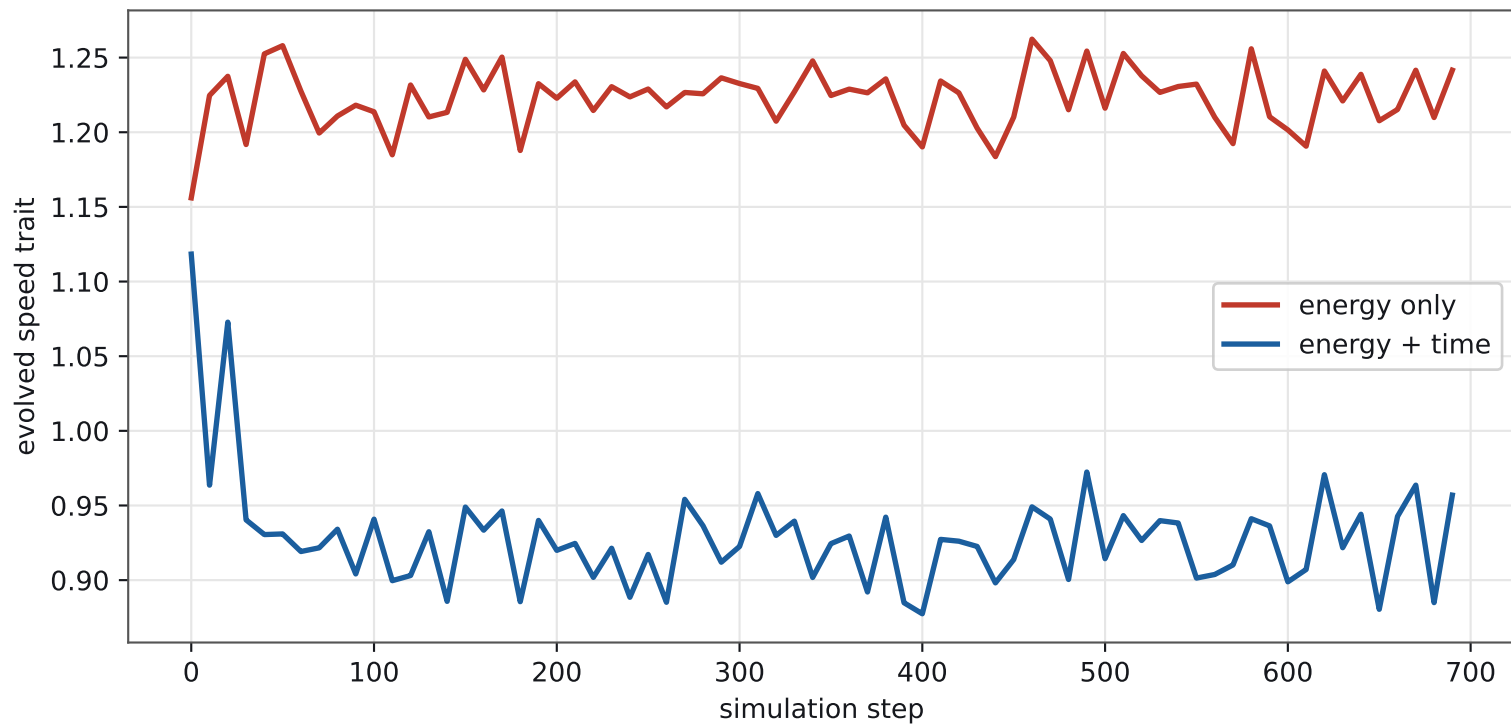
`energy -= move_cost * speed` (default 0.12)

Observed effect

Adding the time law lowers the evolved speed trait relative to the no-cost control (Figure 3): the population shifts toward calmer, more directed movement rather than high-speed wandering. The effect is real but modest at default strength -- it is a refining pressure, not a dominating one.

Open / weak: in a small static world the gain from parsimony is limited; its documented strength in the wider EEC programme comes from selecting GENERALISATION under novelty, which this board does not yet stress. Expect the time law to matter much more once Climate / non-stationarity is co-active.

Figure 3. Time cost selects lower, more efficient speed



5. Perception (cost of looking)

COLOUR: blue AXIS: attention STATUS: real, strong

Definition

A wide field of view is metabolically expensive (cost grows with the square of vision radius).
Organisms are pushed to see only as far as is worth the energy.

Mechanism / parameters

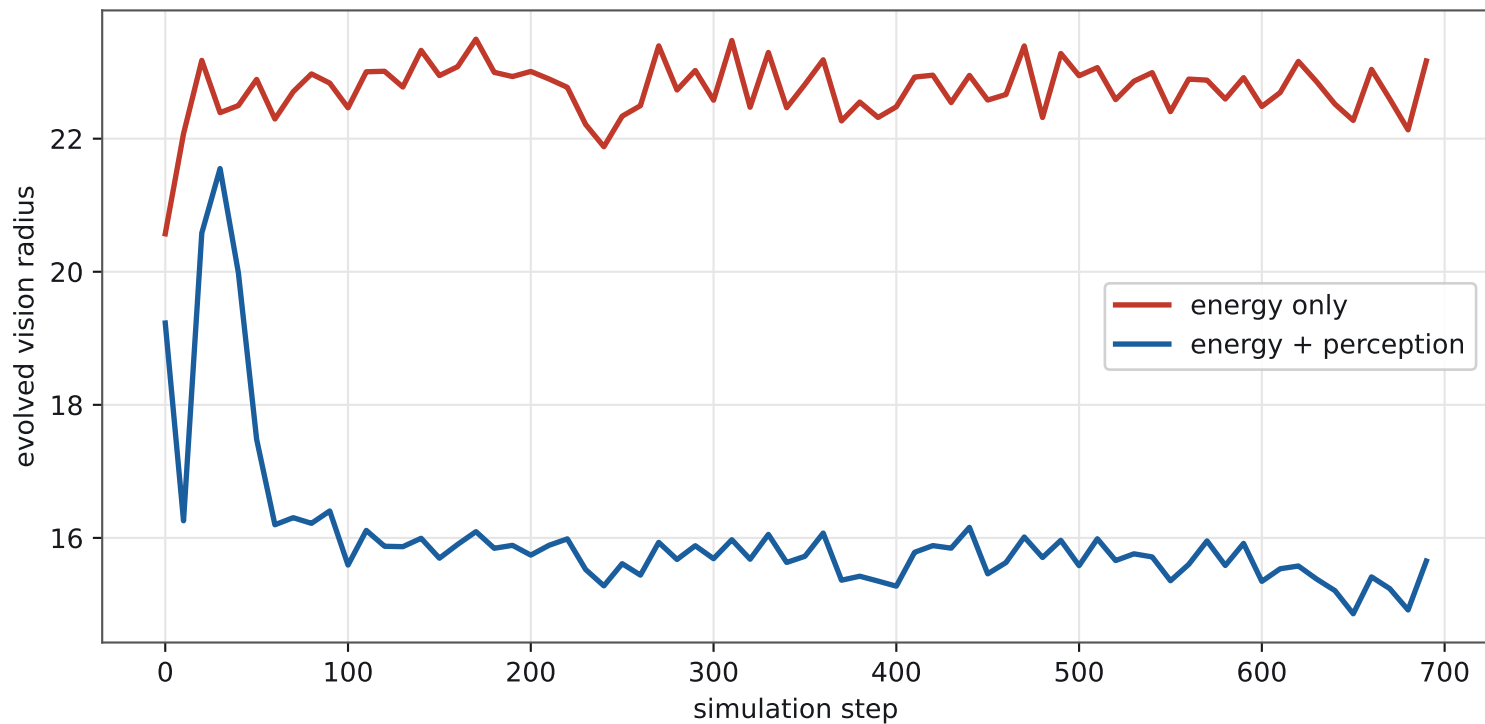
`energy -= cost * vision^2 / 10` (default cost 0.006)

Observed effect

The clearest single-law signature in the system. Vision starts large (random ~20) and, under the perception cost, collapses toward a small economical radius (Figure 4); the no-cost control keeps or grows its vision. In the earlier staged demos this drove vision from ~27 down to ~4.

Open / weak (the 'attention wall'): perception buys an economical FIELD SIZE, but we have not shown it produces SELECTIVE attention -- looking at the right thing rather than simply less. Across the EEC programme, conditional 'what to attend to' required an explicit channel, not just a cost. Cost shrinks the aperture; it does not by itself build a controller for it.

Figure 4. Perception cost shrinks the vision radius



6. Entropy (memory decay)

COLOUR: amber AXIS: active maintenance STATUS: real in programme, weak on this board

Definition

Remembered food locations decay over time; a memory must be refreshed by revisiting or it becomes worthless. Information is not free to hold -- it rots.

Mechanism / parameters

`memory_confidence *= decay` (default 0.90 when on, else 0.995)

Observed effect

Entropy makes stored state a depreciating asset, favouring organisms that periodically re-sense rather than coast on stale memory. In the wider EEC sweeps a decay law produced a measurable premium on active maintenance (recurrent gain ~3.2x).

Open / weak: on the current playground board memory is a single weak channel and the effect is hard to isolate cleanly -- it is the least visually legible of the eight laws here. A dedicated revisit metric and a richer memory representation are needed before claiming a sharp on-board signature. Reported as real-in-principle, under-demonstrated-here.

7. Scarcity (shared / contested food)

COLOUR: yellow AXIS: diversity / niche STATUS: real, verified

Definition

A food patch is finite and is split among everyone feeding on it; a crowd dilutes each bite and depletes the patch. Competition makes proximity costly.

Mechanism / parameters

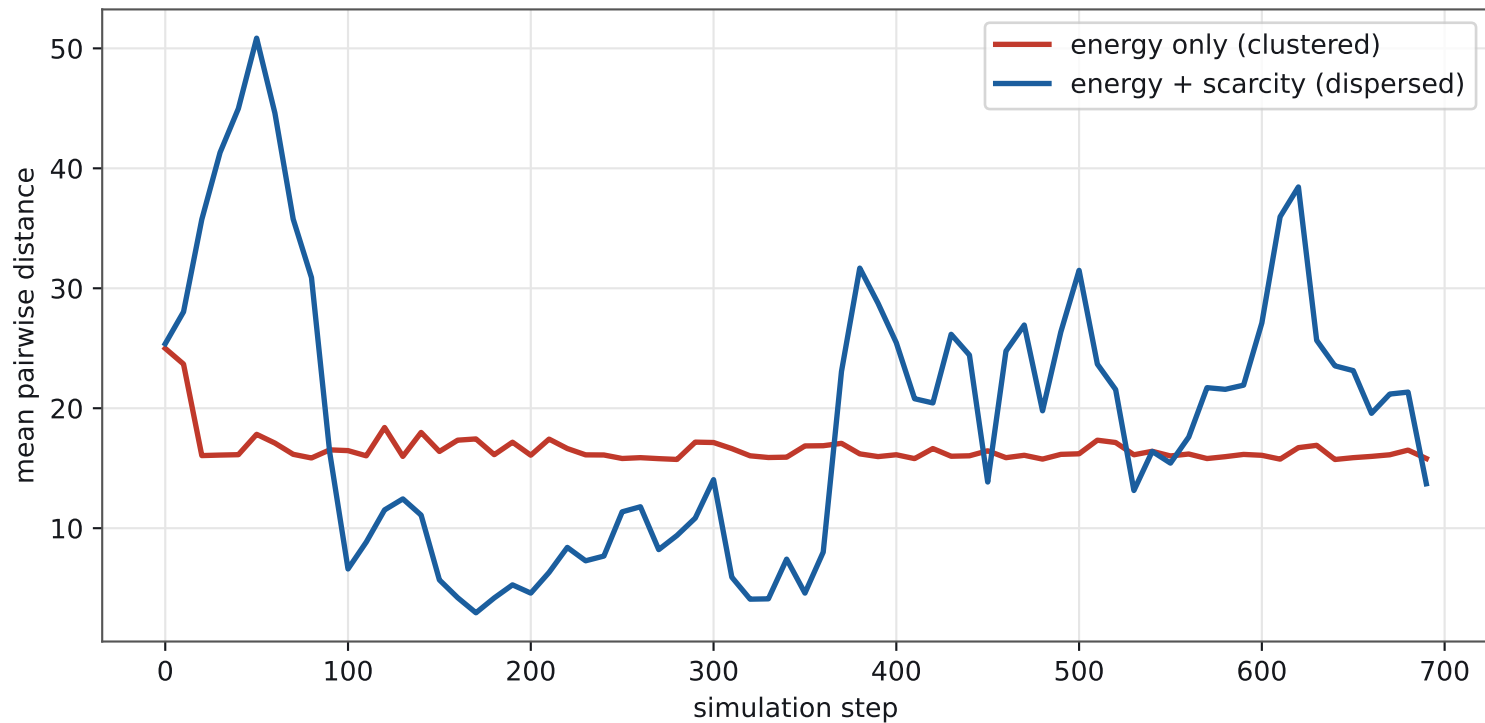
`bite = min(0.5, patch) / (# co-feeders)` ; patch depletes & regrows (regen ~0.004)

Observed effect

Scarcity reliably DISPERSES the population: mean pairwise distance rises and the neighbour-avoidance gene (w_{sep}) is driven positive relative to the abundant-food control (Figure 5). Crowding stops paying, so organisms spread into territory -- the first genuinely social/spatial structure the world produces.

Open / weak: dispersal strength is sensitive to the regen rate and population density; too harsh a scarcity tips into population collapse rather than territory. The 'ecosystem of niches' result from the wider programme needs a population operator that lets sub-populations specialise -- on this board scarcity gives spacing, not yet speciation.

Figure 5. Scarcity disperses the population into territory



8. Communication (signals)

COLOUR: green AXIS: information transfer STATUS: real, but context-dependent

Definition

An organism on food emits a signal; others can sense the nearest signaller and move toward (or away from) it. The genome's w_{sig} sets the sign and strength of the response.

Mechanism / parameters

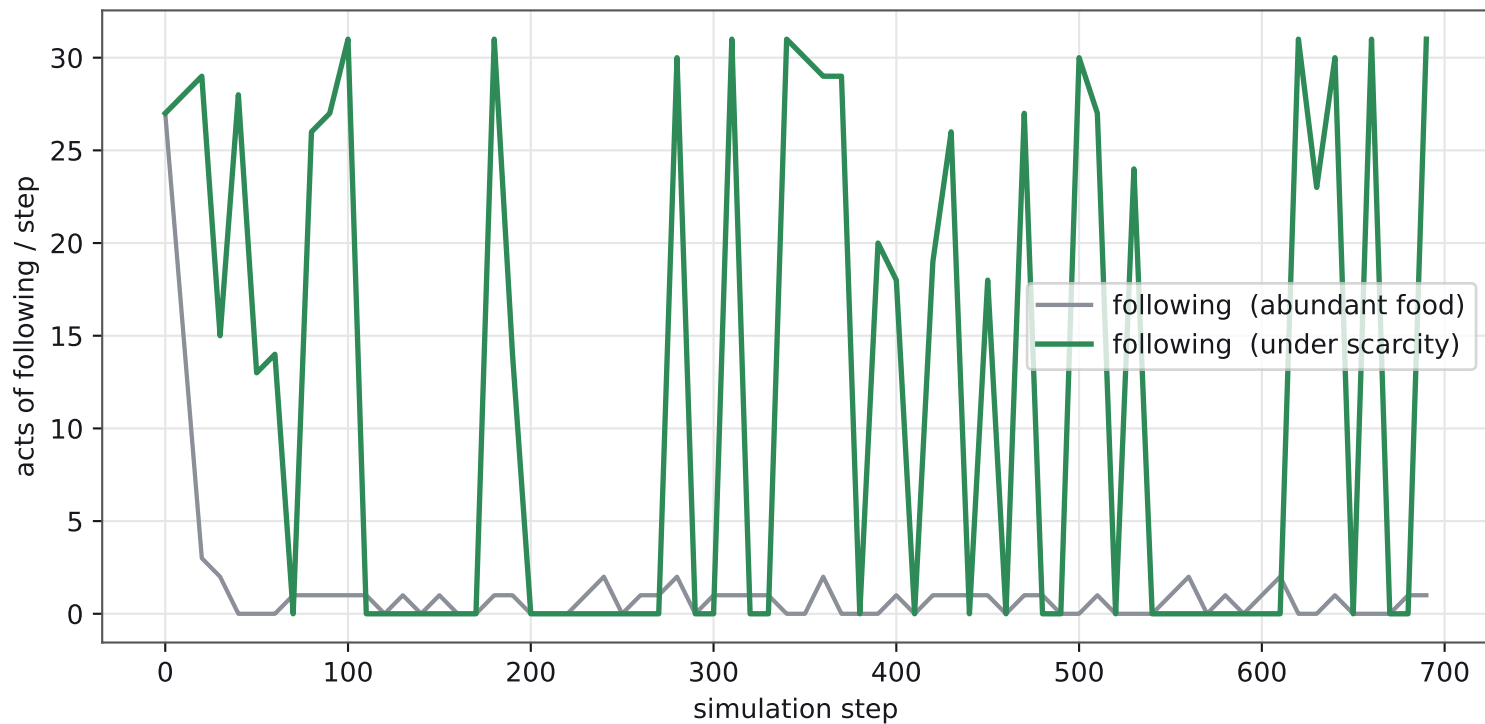
```
signaller = on food ; listener heading +=  $w_{sig}$  * direction(nearest signaller within range)
range default 42
```

Observed effect

Signalling activity tracks how many organisms sit on food; FOLLOWING (acting on a signal) only becomes common when organisms are frequently apart and hungry -- e.g. under scarcity (Figure 6). When food is abundant everyone is already fed and the channel is loud but unused. Crucially, the evolved SIGN of communication is not fixed by this law alone -- it is set by the reproduction regime (Section 11).

Open / weak (the central honest caveat): this is signal-FOLLOWING (attraction), not a symbolic language. Getting an informative, decipherable PROTOCOL to emerge among free-living, individually selected agents runs into the free-rider problem -- a speaker has little private incentive to be honest. In separate experiments a real codebook (MI ~1.5 bits, with dialects) emerged only under COUPLED survival or strong kin structure. On this board, communication is a coordination cue, not yet a vocabulary.

Figure 6. Communication is used only when organisms need each other



9. Climate (rotating season)

COLOUR: cyan AXIS: non-stationarity (spatial) STATUS: mechanically sound, harsh

Definition

The board is quartered and the growing season rotates round-robin through the quadrants. Food has a seedling->mature life-cycle; only MATURE food feeds you, and off-season food exhausts. The resource is non-stationary in space and time.

Mechanism / parameters

```
active = (step // period) % 4 ; maturity grows in active quadrant, decays elsewhere  
see growing food (to migrate) but eat only mature    ; period 140, growth 0.04, decay 0.008
```

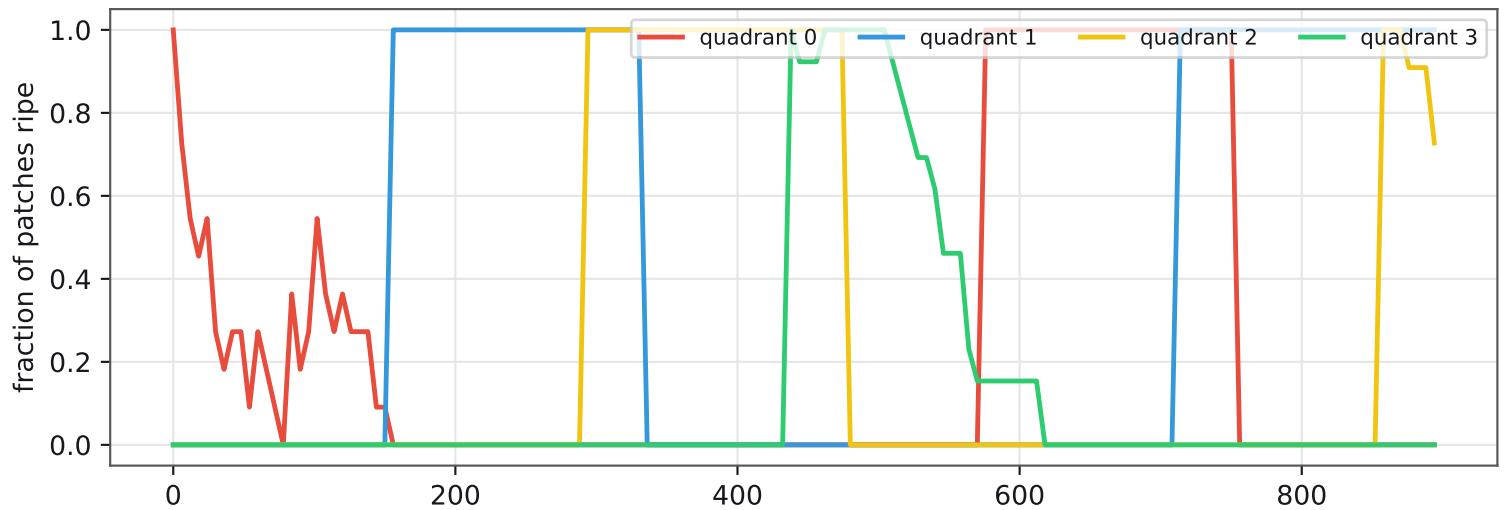
Observed effect

Ripe food cleanly migrates quadrant to quadrant, and the hand-off shows the intended overlap -- the old quadrant exhausts as the next ripens (Figure 7, top). The population is forced to migrate to keep eating: an organism that stays put eats only every fourth season. We verify the herd tracks the bloom above the 25% random baseline, but it LAGS (Figure 7, bottom).

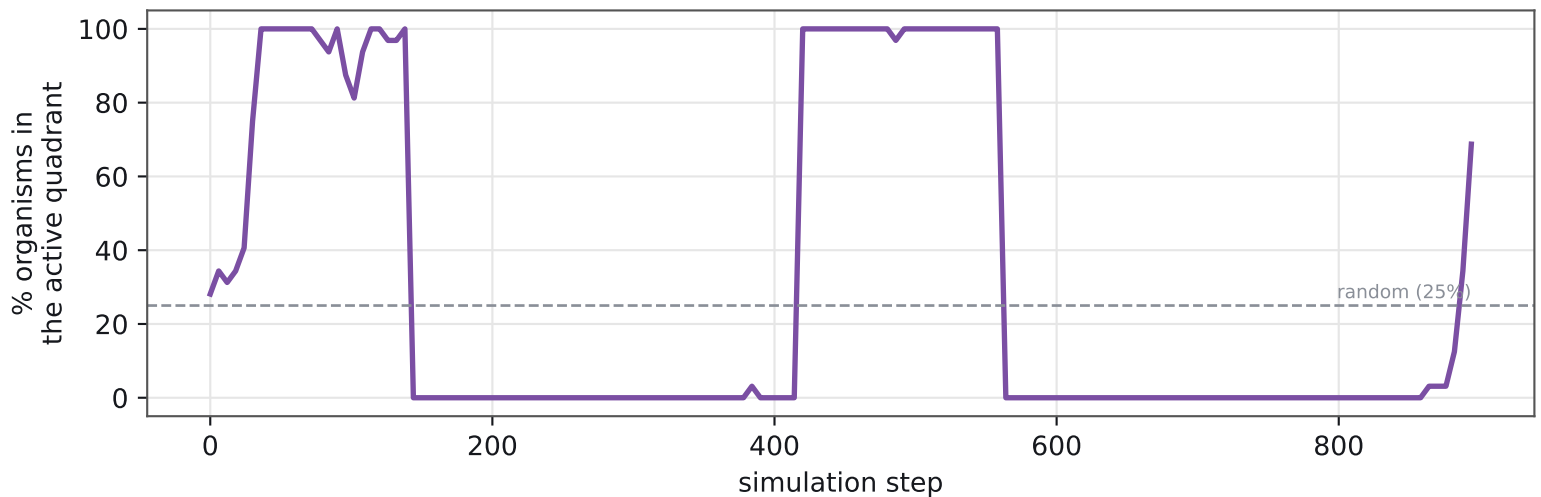
Open / weak: this is the hardest law to keep survivable. Transit time to the next quadrant can exceed starvation time, so the population rides the survival edge and tracks the bloom imperfectly. Making it work required raising the energy cap (reserves for migration), central starts, and even food across quadrants. The survivable parameter window is narrow; outside it the population collapses. We do not yet see clean anticipatory migration (leaving before exhaustion).

Figure 7. Climate — a non-stationary, spatially rotating resource

Rotating growing season: ripe food migrates quadrant to quadrant



Migration tracking (the population lags the bloom)



10. Reproduction (mate-based sex)

COLOUR: pink AXIS: the selection operator itself STATUS: real, with a major interaction

Definition

Turns OFF clone-the-best. An organism now reproduces ONLY with a nearby MATE, and only if BOTH parents are at $\geq 50\%$ of energy capacity. Offspring are a genetic CROSSOVER of the two parents plus mutation; both parents pay an energy cost. Population floats with real births and deaths.

Mechanism / parameters

```
eligible: energy  $\geq 0.5 * \text{capacity}$  ; partner within mate_radius (12)  
child = crossover(parentA, parentB) + mutation ; both pay repro_cost (8) ; cap max_pop (140)
```

Observed effect

Replacing the operator changes the population dynamics from a clamped constant N to a free-floating trajectory bounded by food and the max-pop cap (Figure 8, top). More importantly it changes WHICH genomes can persist -- a lineage must keep its members both well-fed AND sociable enough to meet. This is the lever behind the Section 11 interaction. The 50% energy gate is verified: forcing the population below threshold halts all breeding; above it, breeding resumes.

Open / weak: with scarcity or climate co-active, mate-based reproduction can drive the population to EXTINCTION -- deaths outpace the stricter birth condition. The current mate search is nearest-eligible-within-radius with no mate choice on genome quality (no sexual selection on traits yet), and crossover is uniform per-gene. Population stability across law combinations is not guaranteed and is an active tuning question.

11. Interaction: reproduction shapes communication

The most striking cross-constraint result in the system.

Question

Does the way a population reproduces change what its communication is FOR? We compare the evolved signal-following gene w_{sig} (negative = avoid others, positive = approach others) under asexual cloning vs mate-based sex, holding the world otherwise fixed (energy + communication + perception).

Result

Under cloning, w_{sig} is only weakly constrained and DRIFTS -- across seeds it lands anywhere from strongly anti-social (-0.9) to strongly social (+1.9). Under sexual reproduction it is reliably NON-NEGATIVE on every seed tested (Figure 8, bottom): the strongly anti-social outcomes are eliminated. On the default seed this appears as a clean sign flip (-0.92 -> +0.32).

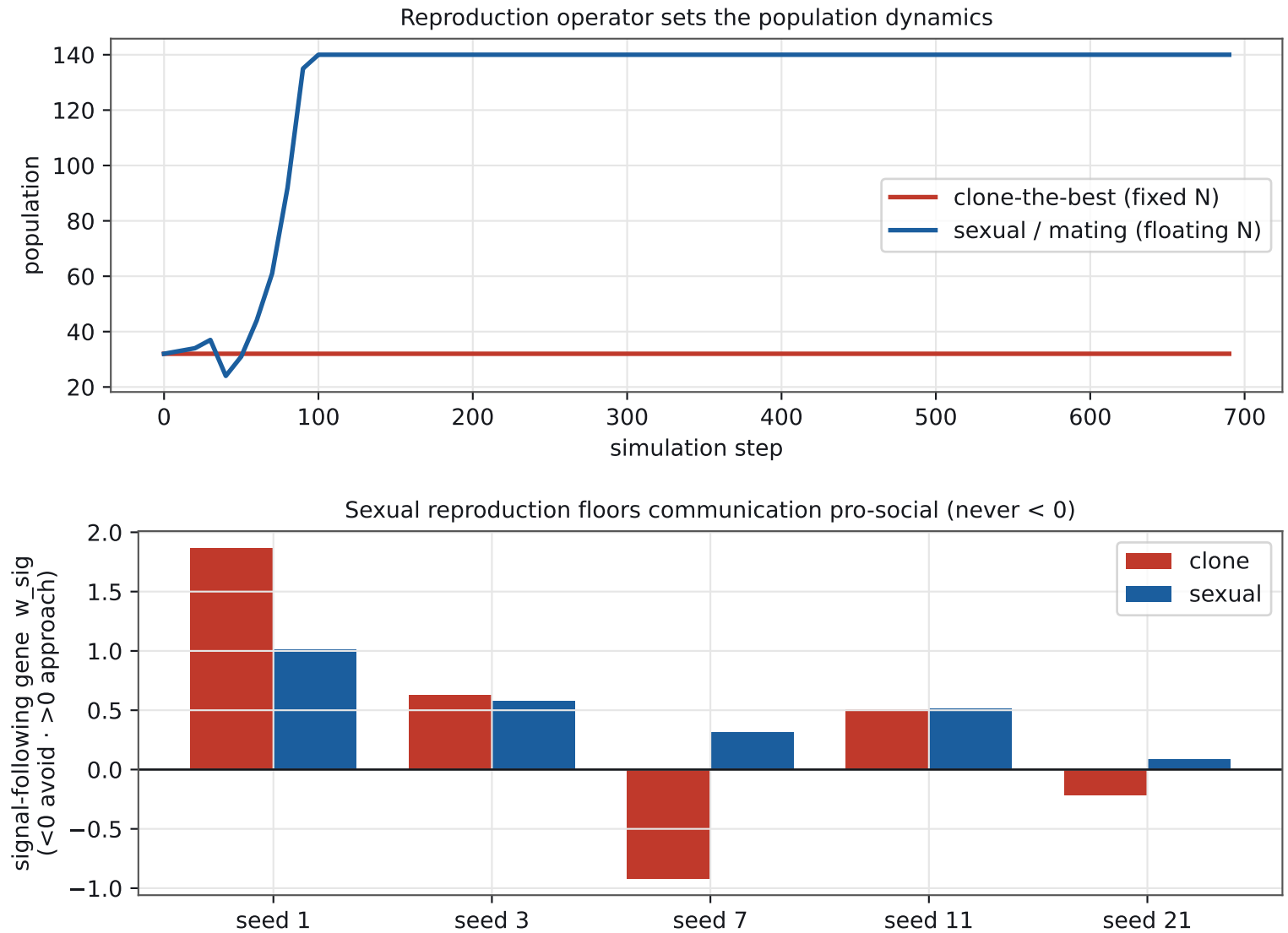
Mechanism

Under cloning, every other organism is purely a competitor for food, so evolution is free to select avoidance. Under sex, every other organism is also a potential MATE and reproduction is impossible without approaching one -- so a lineage that evolves to avoid others simply fails to breed and dies out. Sexual reproduction places a hard FLOOR under social attraction.

Interpretation: the selective VALENCE of a communication channel is not a property of the channel alone -- it is set by the reproduction operator. This mirrors the hypothesised role of sexual selection in the origin of social signalling, and it emerged here from toggling one constraint.

Honesty bound: the dramatic single-seed sign-FLIP is partly seed-specific. The robust, every-seed claim is the weaker-but-solid one: sex eliminates anti-social communication and floors w_{sig} positive. We have not yet measured downstream effects (does the floored sociality raise actual information transfer, or only spatial aggregation?) -- that is the next experiment.

Figure 8. Reproduction — and its grip on the evolution of communication



12. Cross-cutting limitations and open research

Limitations of the instrument

- Minimal organism: a linear 6-gene genome with one fixed behavioural rule. No internal network, no learning within life. Conclusions are about constraint geometry, not realistic ecology.
- Heuristic PO metric: strategy diversity is a genome-dispersion proxy, not a first-principles count of binding constraints. It trends correctly but is not calibrated.
- Fixed-N default operator hides die-off in the 'alive' readout; lifespan is the truer fitness.
- Limited statistics: most single-law figures are one seed. Effects flagged 'mild' or 'weak' have not had confidence intervals computed; treat them as directional.

Constraint-specific open questions

- Perception: cost yields a smaller aperture but not demonstrated SELECTIVE attention (the attention wall).
- Entropy: real in the wider programme, under-demonstrated on this board; needs a revisit metric.
- Communication: signal-following, not a symbolic protocol; honest emergent language needs coupled survival / kin structure, not free-living individual selection.
- Climate: narrow survivable window; no clean anticipatory migration yet.
- Reproduction: extinction risk under scarcity/climate; no mate choice / sexual selection on traits; uniform crossover only.

Active research agenda

1. Quantify the reproduction x communication interaction's downstream effect on real information transfer, not just aggregation.
2. Co-active non-stationarity (Climate) x parsimony (Time) -- does novelty finally make the time law dominant, selecting genuine generalisation?
3. A principled PO estimator (constraint-rank, not genome variance).
4. Mate choice / sexual selection on traits, and population-stability control across law stacks.
5. A richer organism (recurrent internal state) so memory, attention and language have somewhere to live.

Bottom line

The cone is real and the single-law signatures are clean and reproducible. The deepest result -- that the reproduction operator governs the evolution of communication -- is robust in its floored form and striking in its seed-7 form. The honest frontier is everything that needs a richer organism or a population operator with more structure than the current world provides.